Week 0: Preliminary Foundations

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1 Preliminary Knowledge

1.1 Basic Concepts and Types of Options

An option is a financial derivative whose value depends on an underlying asset, most commonly a stock. Unlike holding the stock itself, an option does not represent ownership. Instead, it gives the holder a contractual right. The holder has the right, but not the obligation, to buy or sell the underlying asset at a predetermined strike price K within a specified time period.

There are two fundamental types of options. A call option gives the holder the right to buy the underlying asset at the strike price. A put option gives the holder the right, but not the obligation, to sell the underlying asset at the strike price. Economically, a call benefits from rising prices, while a put benefits from falling prices. The corresponding payoff functions at maturity are

$$\Phi_{\text{call}}(S) = \max(S - K, 0), \quad \Phi_{\text{put}}(S) = \max(K - S, 0).$$

These payoff functions are continuous but not smooth at $S = K$, where a kink appears. This lack of smoothness later influences numerical behavior. Options are further classified by their exercise style. A European option can only be exercised at maturity T. An American option can be exercised at any time **before or at** maturity.

1.2 Key Model Problem: The Black–Scholes Model

To determine the fair value of an option before maturity, we need a model for the underlying asset price. In the classical Black–Scholes framework, the stock price follows a geometric Brownian motion:

$$dS = \mu S dt + \sigma S dW_t,$$

where μ is the drift, σ is the volatility, and W_t is standard Brownian motion. For example, the risk-free asset evolves as $dB = rBdt$, where r is the risk-free interest rate. Then, let the option price be a smooth function $V(S, t)$. Applying Itô's Lemma gives

$$dV = \left(V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right) dt + \sigma S V_S dW_t.$$

By constructing a self-financing portfolio that holds Δ shares of stock and an amount in bonds, and choosing $\Delta = V_S$, the stochastic term disappears. The resulting portfolio must earn the risk-free rate to avoid arbitrage. This leads to the Black–Scholes partial differential equation:

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0.$$

An important feature of this equation is that the drift μ does not appear. The option price depends on volatility and the risk-free rate, but not on the expected return of the stock. This reflects the principle of risk-neutral valuation. Adding appropriate boundary and initial conditions pins down the solution. Different types of options correspond to different boundary conditions, which can increase the complexity of the problem.

1.2.1 European Option and Explicit Solution

For a European option, the PDE is supplemented by the terminal condition

$$V(S,T) = \Phi(S).$$

Plus an appropriate boundary condition, this determines a unique solution. For example, the European call price has the explicit solution [1]

$$C(S,t) = SN(d_1) - Ke^{-r(T-t)}N(d_2),$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t},$$

and $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution. This closed-form formula, however, relies on the fact that early exercise is not allowed.

1.2.2 American Option and the Inequality Constraint

The American option differs fundamentally because it allows early exercise. At every time t before maturity, the holder must decide whether to exercise or continue holding. Therefore, the price must satisfy the inequality

$$V(S,t) \geq \Phi(S).$$

If this condition were violated, arbitrage would exist. One could buy the option and immediately exercise it for a guaranteed profit. Mathematically, the American option is an optimal stopping problem:

$$V(S,t) = \sup_{\tau \in [t,T]} \mathbb{E}^{\mathbb{Q}} \left[e^{-r(\tau - t)} \Phi(S_\tau) \right].$$

The formula means that the option value $V(S,t)$ equals the maximum over all possible stopping times $\tau \in [t,T]$. For each possible exercise time τ, we compute the discounted payoff $e^{-r(\tau-t)}\Phi(S_\tau)$, take its expectation under the risk-neutral measure $\mathbb{Q}$, and then choose the stopping time that gives the highest value. The PDE characterization becomes a variational inequality:

$$\min(-\mathcal{L}V, V - \Phi) = 0,$$

where

$$\mathcal{L}V = V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV.$$

This means that

- In the continuation region ($\mathcal{L}V = 0$, $V > \Phi$), where holding is optimal, the Black–Scholes PDE holds.
- In the exercise region, the value equals the payoff ($V = \Phi$, $\mathcal{L}V < 0$). The boundary separating these two regions is unknown in advance. This is a free-boundary problem.

Note that the American call option without dividends simplifies the analysis since if the underlying asset pays no dividends, early exercise of a call option is never optimal. Thus American Call = European Call. Thus, **the early exercise issue mainly affects American puts or dividend-paying assets**. The standard BS equation assumes that the price V only depends on one spatial variable S. In practice, many derivatives depend on multiple risk factors. For d underlying assets, the value function becomes

$$V = V(S_1, S_2, \ldots, S_d, t),$$

and the PDE takes the general form

$$V_t + \sum_{i=1}^{d} rS_i V_{S_i} + \frac{1}{2} \sum_{i,j=1}^{d} a_{ij}(S) V_{S_i S_j} = rV.$$

Models with stochastic volatility add additional state variables. In calibration tasks, one must repeatedly solve the PDE for different parameter combinations (r, σ, q, T, K). For finite difference methods, the computational complexity scales approximately as $O(N^d)$, where N is the number of grid points per dimension. As d increases, the total number of grid points grows exponentially. This is known as the curse of dimensionality. In low dimensions, classical numerical methods such as finite difference schemes are accurate and efficient. However, as the dimension increases, grid-based methods become computationally expensive and memory-intensive. Scientific machine learning methods, including Physics-Informed Neural Networks (PINNs), offer a different approach. Instead

of constructing a full tensor grid, a neural network approximates the solution function directly. The PDE is enforced through a loss function evaluated at sampled points in the domain.

2 Literature Review

2.1 Classical Numerical Methods

The classical numerical schemes can be broadly grouped into two main categories.

- Tree and simulation methods.
  - Binomial tree models approximate the underlying asset dynamics by a discrete-time recombining lattice and compute the option value via backward induction, naturally incorporating early exercise.
  - Monte Carlo methods simulate a large number of stochastic paths and estimate discounted payoffs by averaging. While highly flexible and well-suited for high-dimensional European options, they require additional techniques, such as regression-based continuation value approximation, to handle the optimal stopping feature of American contracts.
- PDE-based mesh methods. These approaches exploit the partial differential equation representation of option prices under the Black–Scholes framework. The computational domain is discretized in space and time, and derivatives are approximated numerically. The mesh-based methods typically follow two steps:
  1. The generation of grid points or elements through spatial and temporal discretization.
  2. The solution of the resulting discrete algebraic equations, often by using iterative algorithms.

  The widely used realizations include:
  - Finite Difference Methods (FDM), which directly approximate derivatives by difference quotients.
  - Finite Element Methods (FEM), which use a variational formulation, dividing the domain into elements, approximating locally, and assembling a global solution [7].
  - Finite Volume Methods (FVM), which enforce conservation principles over control volumes and are particularly suited for flux-based formulations.

Among mesh-based approaches, finite difference methods have historically been the most widely used in financial engineering due to their simplicity and direct applicability to parabolic option pricing equations. In the American setting, spatial discretization typically leads to a linear complementarity problem (LCP), which must be solved at each time step. We now focus on this framework and organize the discussion according to how the early-exercise constraint, equivalently the free-boundary condition, is treated within the discrete system.

2.1.1 Finite Difference Discretization and LCP Formulation

Under the Black–Scholes framework, the American option price satisfies the variational inequality

$$\min\left(\partial_t V + \mathcal{L}V - rV,\ V - \Phi\right) = 0,$$

where

$$\mathcal{L}V = \frac{1}{2}\sigma^2 S^2 \partial_{SS} V + rS\partial_S V.$$

Early finite difference treatments were developed by [2], which discretized the PDE and enforced early exercise via recursive comparison. Using central differences,

$$\partial_S V_i \approx \frac{V_{i+1} - V_{i-1}}{2\Delta S}, \qquad \partial_{SS} V_i \approx \frac{V_{i+1} - 2V_i + V_{i-1}}{\Delta S^2},$$

together with implicit time stepping, the semi-discrete system becomes

$$AV^{n+1} = b.$$

For American options, this system must satisfy the complementarity constraint

$$\min(AV^{n+1} - b,\ V^{n+1} - \Phi) = 0,$$

which yields an LCP. The inequality constraint transforms the linear system into a constrained algebraic problem.

2.1.2 LCP Solvers and Complementarity Enforcement

Once discretized, the problem becomes

$$V \geq \Phi, \quad AV - b \geq 0, \quad (V - \Phi)^T (AV - b) = 0. \tag{1}$$

One of the earliest projected relaxation approaches was developed by Cryer [4], forming the basis of the Projected Successive Over-Relaxation (PSOR) method. The iteration reads

$$V_i^{(k+1)} = \max\left(\Phi_i, V_i^{(k)} + \omega \frac{b_i - (AV^{(k)})_i}{A_{ii}}\right),$$

which enforces complementarity pointwise and remains effective in one spatial dimension.

An alternative strategy is the penalty method, analyzed for American options in [5]. The penalized formulation replaces the inequality by

$$-\mathcal{L}V + \frac{1}{\varepsilon}\max(\Phi - V, 0) = 0,$$

where constraint violations introduce a forcing term. As $\varepsilon \to 0$, the solution converges to that of the variational inequality.

2.1.3 Free-Boundary Reformulations

Instead of encoding early exercise through complementarity shown in (1), one may explicitly track the optimal exercise boundary $S^*(t)$. In the continuation region,

$$\partial_t V + \mathcal{L}V - rV = 0,$$

subject to $V(S^*(t), t) = \Phi(S^*(t))$, $\partial_S V(S^*(t), t) = \partial_S \Phi(S^*(t))$. Front-fixing transformations map the moving boundary to a fixed spatial domain, for example, via

$$x = \frac{S}{S^*(t)}.$$

This converts the free-boundary problem into a nonlinear PDE on a fixed domain. Such approaches provide accurate boundary approximations and clear economic interpretation, but introduce additional nonlinear coupling.

2.1.4 High-Dimensional Acceleration and Operator Splitting

For multi-factor models such as the Heston model, the governing PDE takes the form

$$V_t = L_S V + L_v V.$$

Direct discretization leads to large, sparse systems. Alternating Direction Implicit (ADI) schemes split the multidimensional operator into one-dimensional substeps:

$$(I - \theta \Delta t L_S)V^* = \text{rhs},$$
$$(I - \theta \Delta t L_v)V^{n+1} = V^*.$$

This substantially reduces computational cost while retaining stability. [8] introduced operator splitting combined with projection for American options, separating the PDE step from the obstacle enforcement via $V^{n+1} = \max(V^*, \Phi)$. This decoupling makes multidimensional American option pricing computationally feasible.

2.2 Neural and Scientific Machine Learning Approaches

All the previously discussed approaches are mesh-based methods, as they rely on the construction of a spatial–temporal grid and the solution of a discrete algebraic system. However, mesh generation may become computationally expensive or even infeasible in high-dimensional problems. Moreover, discretization may introduce approximation bias between the continuous PDE formulation and its discrete counterpart. To overcome these limitations, recent research has explored meshless and Deep Learning-based strategies for solving PDEs arising in finance. In particular, neural networks have been employed as universal function approximators to solve high-dimensional PDEs without explicit grid construction. (e.g., DGM [10], deep BSDE [3]). These approaches are especially attractive in high-dimensional settings, where classical mesh-based methods suffer from the curse of dimensionality.

Within this context, PINNs introduced in [9], provide a general framework for solving forward and inverse PDE problems by embedding the governing equation and boundary conditions directly into the loss function:

$$V(S,t) \approx V_\theta(S,t),$$

where V_θ is represented by a neural network with trainable parameters θ. The loss function incorporates the PDE residual together with initial and boundary conditions evaluated at collocation points. For the Black–Scholes equation, the residual-based loss typically takes the form

$$\mathcal{L}_{\mathrm{PDE}}(\theta) = \left\| \partial_t V_\theta + \mathcal{L} V_\theta - r V_\theta \right\|^2,$$

with additional penalty terms enforcing payoff and boundary conditions. A typical example is [13]. Extensions to American options are considerably more challenging due to the presence of a free boundary and complementarity constraints, which transform the Black–Scholes equation into a nonlinear variational inequality. In the context of American put pricing, [6] formulate a PINN approach that explicitly incorporates value-matching and smooth-pasting conditions and propose a stabilized training strategy for the free boundary, demonstrating improved convergence in both single- and multi-asset settings. Subsequent work extends the framework to more complex dynamics. [11] address fractional Black–Scholes free-boundary problems using a combined PINN architecture. Under jump-diffusion dynamics, [12] embeds the Merton PIDE and early-exercise penalty into the network objective and provides convergence analysis under data-scarce conditions.

2.3 Evaluation Metrics and Comparative Criteria

To rigorously compare methods, several metrics are essential.

- Price accuracy is measured by

$$\|V - V^{ref}\|_{L^\infty}$$

which captures the worst-case error, and

$$\|V - V^{ref}\|_{L^2}$$

which measures average deviation across the domain.

- Greek accuracy evaluates sensitivity errors:

$$\|\partial_S V - \partial_S V^{ref}\|,$$

since mispricing of deltas can lead to incorrect hedging.

- Complementarity violation is quantified by

$$\|\max(\Phi - V, 0)\|.$$

This metric evaluates whether the numerical solution respects the American constraint.

- Free-boundary accuracy is measured via

$$|S^*(t) - S^{*,ref}(t)|.$$

Even if price errors are small, inaccurate boundary prediction may indicate structural inconsistency.

3 Implementation Plan

3.1 Phase 1: 1D European Put – Building a Stable PDE Solver

The first step is to implement a standard 1D European put solver under the Black–Scholes model. This stage serves two purposes:

- Verify that the spatial discretization and time stepping scheme are correct.
- Create a debugging baseline before adding inequality constraints.

The PDE is

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0,$$

with terminal condition

$$V(S, T) = \max(K - S, 0).$$

Implementation details:

- Define computational domain $S \in [0, S_{\max}]$, typically $S_{\max} = 4K$.

- Use $N_S = 200$ spatial grid points.
- Use $N_T = 200$ time steps.
- Apply Crank–Nicolson time discretization.
- Use Rannacher smoothing for the first two time steps.

At this stage, the numerical solution will be compared against the closed-form Black–Scholes price. We will compute

$$\|V - V^{BS}\|_{L^\infty}$$

to confirm convergence. This phase ensures that discretization, boundary conditions, and matrix assembly are implemented correctly.

3.2 Phase 2: 1D American Put – Penalty Formulation

Once the European solver is validated, we introduce early exercise using the penalty approach. The American variational inequality

$$\min(-\mathcal{L}V, V - \Phi) = 0$$

is approximated by the penalized PDE

$$-\mathcal{L}V + \frac{1}{\varepsilon}\max(\Phi - V, 0) = 0.$$

This converts the inequality into a nonlinear PDE. Implementation tasks:

- Keep same spatial and temporal grid as Phase 1.
- Choose $\varepsilon = 10^{-4}$ initially.
- Solve nonlinear system at each time step via Newton iteration.
- Monitor convergence of Newton residual.

We will also extract the discrete exercise boundary:

$$S^*(t_n) = \min\{S_i : V_i^n > \Phi_i\}.$$

This phase provides a stable, high-accuracy reference for the American price and free boundary.

3.3 Phase 3: Classical LCP Solver via PSOR

To compare formulations, we implement the strict complementarity approach. The discrete system satisfies

$$V \geq \Phi, \quad AV - b \geq 0, \quad (V - \Phi)^T (AV - b) = 0.$$

We solve this using PSOR:

$$V_i^{k+1} = \max\left(\Phi_i, V_i^k + \omega \frac{b_i - (AV^k)_i}{A_{ii}}\right).$$

Implementation plan:

- Use relaxation parameter $\omega = 1.5$.
- Stopping tolerance 10^{-8}.
- Record iteration counts per time step.

This phase allows us to compare penalty vs. exact complementarity enforcement in terms of convergence speed and accuracy.

3.4 Phase 4: Evaluation Metrics for Fair Comparison

To compare numerical and neural methods consistently, we compute the metrics in section 2.3. Runtime and iteration counts will also be recorded.

3.5 Phase 5: PINN for 1D European Put

Before addressing the American constraint, we implement a PINN for the European PDE. Network architecture:

- 5 hidden layers
- 64 neurons per layer
- activation: tanh

Loss function:

$$J(\theta) = \mathbb{E}_{(S,t)} |\mathcal{L} V_\theta|^2 + \mathbb{E}_S |V_\theta(S, T) - \Phi(S)|^2.$$

Training:

- 10,000 collocation points
- Adam optimizer (learning rate 10^{-3})
- 5,000–10,000 epochs

We verify that the PINN reproduces the European solution within acceptable error bounds.

3.6 Phase 6: PINN for American Put

The American constraint is incorporated via obstacle penalty:

$$J_{ineq} = \lambda \mathbb{E}(\max(\Phi - V_\theta, 0)^2).$$

Total loss:

$$J_{total} = J_{PDE} + J_{terminal} + J_{ineq}.$$

Implementation steps:

- Start with $\lambda = 10$.
- Increase collocation points to 20,000.
- Monitor constraint violation separately.
- Compare against FDM reference.

Optional improvement: Introduce augmented Lagrangian or residual adaptive refinement after baseline PINN is validated.

3.7 Expected Progression

The implementation proceeds in increasing difficulty:

1. European FDM (linear, stable).
2. American penalty FDM (nonlinear).
3. American PSOR (exact complementarity).
4. European PINN.
5. American PINN with obstacle penalty.

This staged progression ensures correctness at each level before increasing model complexity.

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